

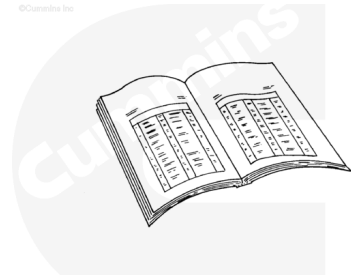
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- Drain the cooling system. Use the following procedure for ISM, ISMe, and QSM11 engines. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Drain the cooling system. Use the following procedure for Signature™, ISX, and QSX15 engines. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
- Drain the cooling system. Use the following procedure for ISX15 CM2250 SN engines. Refer to Procedure 008-018 in

Section 8. (/qs3/pubsys2/xml/en/procedures/175/175-008-018-tr.html)

- Remove the air inlet and outlet connections from the air compressor. Use the following procedure for ISM, ISMe, and QSM11 engines. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/35/35-012-014-tr.html)
- Remove the air inlet and outlet connections from the air compressor. Use the following procedure for Signature™, ISX, and QSX15 engines. Refer to Procedure 012-014 in Section 12 (/qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html)
- Remove the air inlet and outlet connections from the air compressor. Use the following procedure for ISX15 CM2250 SN engines. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/175/175-012-014-tr.html)
- Remove the air compressor coolant lines. Use the following procedure for ISM, ISMe, and QSM11 engines. Refer to Procedure 012-004 in Section 12 . (/qs3/pubsys2/xml/en/procedures/35/35-012-004-tr.html)
- Remove the air compressor coolant lines. Use the following procedure for Signature™, ISX, and QSX15 engines. Refer to Procedure 012-004 in Section 12. (/qs3/pubsys2/xml/en/procedures/10/10-012-004-tr.html)
- Remove the air compressor coolant lines. Use the following procedure for ISX15 CM2250 SN engines. For Refer to Procedure 012-004 in Section 12. (/qs3/pubsys2/xml/en/procedures/175/175-012-004-tr.html)
- Remove the air governor or air governor signal line, if equipped. Refer to the OEM service manual.

Remove

Remove the six air compressor cylinder head mounting capscrews.

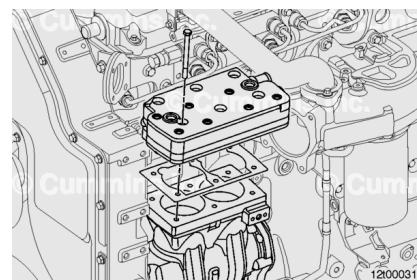
Discard the capscrews.

Note : Record the position of the intake and exhaust ports in the air compressor cylinder head with respect

to the crankcase for correct assembly.

Remove the air compressor head and gasket.

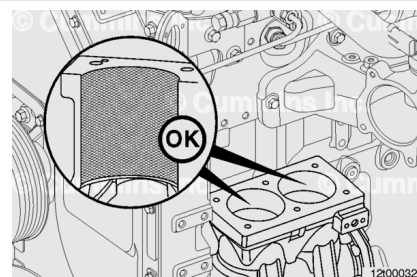
Discard the gasket.



Inspect for Reuse

⚠ CAUTION ⚠

Do not use an abrasive cleaning tool on any surface of the air compressor cylinder head. If an abrasive cleaning tool is used, the air compressor cylinder head can become damaged.

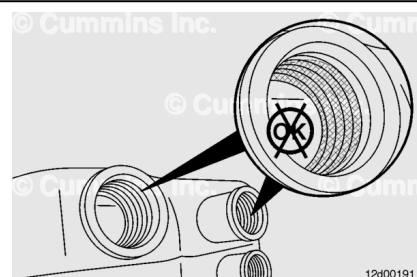


Inspect the air compressor cylinder bores for damage. Use the following procedure for ISM, ISMe, and QSM11 engines. Refer to Procedure 012-014 in Section 12 (</qs3/pubsys2/xml/en/procedures/35/35-012-014-tr.html>).

Inspect the air compressor cylinder bores for damage. Use the following procedure for Signature™, ISX, and QSX15 engines. Refer to Procedure 012-014 in Section 12. (</qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html>)

Inspect the air compressor cylinder bores for damage. Use the following procedure for ISX15 CM2250 SN engines. Refer to Procedure 012-014 in Section 12. (</qs3/pubsys2/xml/en/procedures/175/175-012-014-tr.html>)

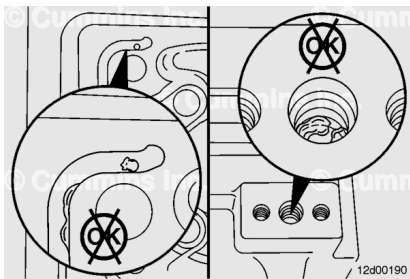
Inspect the intake and exhaust ports for excessive carbon buildup. Refer to Procedure 012-003 in Section 12. (</qs3/pubsys2/xml/en/procedures/101/101-012-003-tr.html>)



Inspect the air governor signal channel in the air compressor cylinder head. The channel **must** appear as a clean aluminum surface.

The air compressor cylinder head **must** be replaced if there is any sign of rust, dirt, or exhaust gas recirculation debris present.

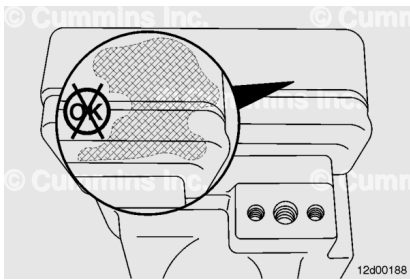
Note : If rust is found in the air governor signal channel, it is the result of poor system maintenance. Check for proper operation of the air dryer and the wet tanks. Refer to the OEM service manual.



Inspect the air compressor cylinder head for signs of a coolant leak.

A coolant leak will be identified by an area of engine paint appearing to have been washed away. White crystalline clusters of material are **not** a sign of a coolant leak.

If a leak is found, replace the air compressor cylinder head.

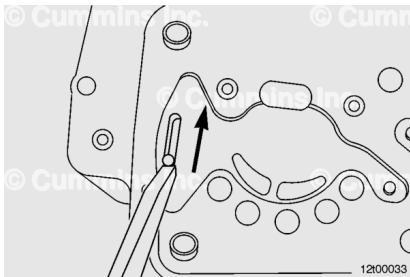


Verify the unloader slider plate is fully closed when at rest.

Remove the unloader slider plate and verify proper unloader piston travel. Use a stiff wooden handle or equivalent non-abrasive device to move the unloader piston by pushing back on the unloader slider plate guide pin in the direction of arrow as indicated. Be careful **not** to damage the aluminum surface of the head or drop debris into the open slot in which the guide pin travels.

Note : When released, the unloader piston should return to the loaded position with a quick, smooth motion and an audible snap.

If the unloader piston does **not** return to the loaded position with a quick, smooth motion and audible snap, reference to the table below for repair direction:



Guide Pin Action	Repair Direction
Guide pin can be moved to loaded (closed) position by hand.	Spring is broken or weak. Replace the cylinder head.
Guide pin does not return or slow to return to loaded position.	Refer to Procedure 012-003 in Section 12. (/qs3/pubsys2/xml/en/procedures/101/101-012-003-tr.html)

Assemble

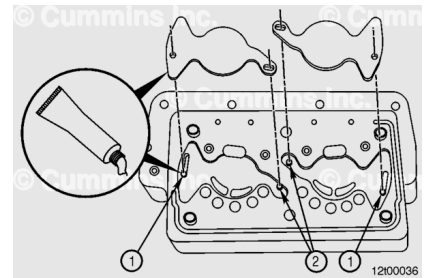
Note : The assembly process is not required for new air compressor cylinder head kits. These kits include a cylinder head with all of the gaskets assembled and ready for installation on the air compressor crankcase.

Note : Make sure the unloader slider plate, gasket surfaces and valve plate are free of any debris before assembling the unloader slider plate.

Apply a spot of assembly lubricant, Part Number 3163087, or equivalent, to the back of the unloader slider plates and to the unloader slider plate guide pins (1).

Install the unloader slider plates onto the air compressor cylinder head by locating it on the unloader slider plate pivot pins (2) and the unloader slider plate guide pins (1).

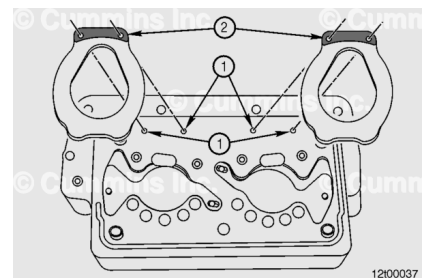
Note : Make sure the unloader slider plates are properly seated and that enough assembly lubricant has been used to securely hold them in position during the installation procedure, or damage to the unloader slider plates can occur.



Install the air compressor intake reed valve. Align the two holes in each reed valve with the two pins in the air compressor cylinder head (1).

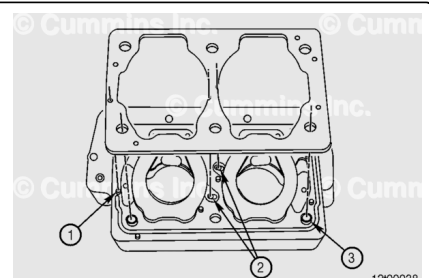
The free end of each intake reed valve will cover one of the two sets of five circular intake holes in the bottom of the air compressor cylinder head when properly installed.

The gasket material on the reed valve (2) **must** be exposed when properly installed.



Install the air compressor cylinder head gasket. Make sure the air compressor cylinder head gasket is oriented so the governor signal channel of the cylinder head is exposed by five holes (1). The two unloader slider pivot pins will be exposed when the gasket is properly installed (2).

The air compressor cylinder head gasket **must** be located on the four locating spring pins (3) protruding from the bottom of the air compressor cylinder head. The air compressor cylinder



head gasket will allow free movement of the air compressor intake reed valve when properly installed.

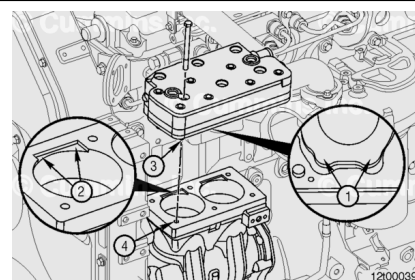
Note : The air compressor cylinder head gasket will not allow proper movement of the intake reed valve and will prevent the air compressor from building air pressure properly when improperly installed.

Install

Place the air compressor cylinder head on the air compressor.

The free ends of the reed valves (1) will align with the chamfers (2) on the air compressor cylinders.

The four locating pins (3) will align with four of the holes (4) in the air compressor and the head will drop into place.



Note : Use new gaskets and capscrews when installing a used air compressor cylinder head. These components are not reusable and are included in the gasket and seals kit. Reference the Assembly section above for proper assembly of the necessary components.

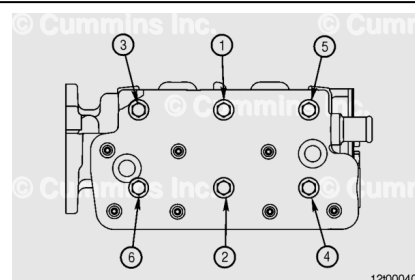
Place the air compressor cylinder head on the air compressor. The free ends of the reed valves will align with the chamfers on the air compressor cylinders. The four locating pins will align with four of the holes in the air compressor and the head will drop into place.

Install the six air compressor cylinder head mounting capscrews.

Tighten the capscrews in the sequence illustrated. Use the torque plus angle method.

Torque Value:

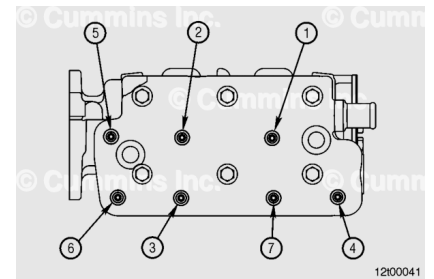
1. 25 n•m [221 in-lb]
2. Rotate capscrew numbers 1 and 2 by 150 degrees
3. Rotate capscrew numbers 3, 4, 5, and 6 by 120 degrees.



Tighten the seven Torx™ head capscrews in the sequenced illustrated.

Torque Value:

1. 6 n·m [53 in-lb]
2. Rotate 135 degrees



Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the air compressor coolant lines. Use the following procedure for ISM, ISMe, and QSM11 engines. Refer to Procedure 012-004 in Section 12 (</qs3/pubsys2/xml/en/procedures/35/35-012-004-tr.html>).
- Install the air compressor coolant lines. Use the following procedure for Signature™, ISX, and QSX15 engines. Refer to Procedure 012-004 in Section 12. (</qs3/pubsys2/xml/en/procedures/10/10-012-004-tr.html>)
- Install the air compressor coolant lines. Use the following procedure for ISX15 CM2250 SN engines. Refer to Procedure 012-004 in Section 12. (</qs3/pubsys2/xml/en/procedures/175/175-012-004-tr.html>)
- Install the air inlet and outlet connections to the air compressor. Use the following procedure for ISM, ISMe, and QSM11 engines. Refer to Procedure 012-014 in

Section 12 (/qs3/pubsys2/xml/en/procedures/35/35-012-014-tr.html).

- Install the air inlet and outlet connections to the air compressor. Use the following procedure for Signature™, ISX, and QSX15 engines. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html)
- Install the air inlet and outlet connections to the air compressor. Use the following procedure for ISX15 CM2250 SN engines. Refer to Procedure 012-014 in Section 12. (/qs3/pubsys2/xml/en/procedures/175/175-012-014-tr.html)
- Install the air compressor governor or governor signal line. Refer to the OEM service manual.
- Fill the cooling system. Use the following procedure for ISM, ISMe, and QSM11 engines. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html).
- Fill the cooling system. Use the following procedure for Signature™, ISX, and QSX15 engines. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
- Fill the cooling system. Use the following procedure for ISX15 CM2250 SN engines. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/175/175-008-018-tr.html)
- Connect the batteries. Refer to the OEM service manual.
- Operate the engine to normal operating temperature and check for leaks.